

Question 1

(a) Given $r(x, y) = \sqrt{x^2 + y^2} = (x^2 + y^2)^{\frac{1}{2}}$

$$\begin{aligned}
 r_x &= \frac{1}{2}(x^2 + y^2)^{-\frac{1}{2}} \cdot 2x \\
 &= x(x^2 + y^2)^{-\frac{1}{2}} \\
 &= \frac{x}{r} \\
 r_{xx} &= \frac{1}{r} - \frac{1}{2}x(x^2 + y^2)^{-\frac{3}{2}} \cdot 2x \\
 &= \frac{1}{r} - x^2(x^2 + y^2)^{-\frac{3}{2}} \\
 &= \frac{1}{r} - \frac{x^2}{r^3} \\
 &= \frac{r^2 - x^2}{r^3} \\
 &= \frac{y^2}{r^3} \\
 r_y &= \frac{y}{r} \\
 r_{yy} &= \frac{x^2}{r^3}
 \end{aligned}$$

(b) Given $u(x, y) = f(r(x, y))$

$$\nabla^2 u = u_{xx} + u_{yy}$$

Find f_{xx} and f_{yy}

$$\begin{aligned}
 f_x &= f'(r)r_x \\
 f_{xx} &= f''(r)r_x + f'(r)r_{xx} \\
 &= f''(r)\frac{x}{r} + f'(r)\frac{y^2}{r^3} \\
 &= \frac{1}{r} \left(f''(r)x + f'(r)\frac{y^2}{x^2 + y^2} \right) \\
 f_{yy} &= \frac{1}{r} \left(f''(r)y + f'(r)\frac{x^2}{x^2 + y^2} \right)
 \end{aligned}$$

So we have

$$\begin{aligned}\nabla^2 u &= \frac{1}{r} \left(f''(r)x + f'(r)\frac{y^2}{x^2 + y^2} + f''(r)y + f'(r)\frac{x^2}{x^2 + y^2} \right) \\ &= \frac{1}{r} (f''(r)(x + y) + f'(r)) \\ &= f''(r)\frac{x + y}{r} + \frac{1}{r}f'(r)\end{aligned}$$

(c) So we have the ODE $f''(r) + \frac{1}{r}f'(r) = 0$

$$\begin{aligned}f''(r) &= -\frac{1}{r}f'(r) \\ \frac{d}{dr}f'(r) &= -\frac{1}{r}f'(r)\end{aligned}$$

Let $u = f'(r)$

$$\begin{aligned}\frac{du}{dr} &= -\frac{1}{r}u \\ \frac{1}{u}du &= -\frac{1}{r}dr \\ \ln u &= -\ln r + c \\ u &= \frac{C}{r} \\ f'(r) &= \frac{C}{r} \\ \frac{d}{dr}f(r) &= \frac{C}{r} \\ f(r) &= C \ln |r| + B\end{aligned}$$

Question 2

- (a) Denote t as y . The wave equation becomes $u_{yy} - c^2 u_{xx} = 0$. We have the general form of a constant-coefficients PDE as

$$Au_{xx} + Bu_{xy} + Cu_{yy} + F(x, y, u, u_x, u_y) = 0$$

In this case, we have $A = -c^2, B = 0, C = 1$, so our discriminant is

$$D = 0^2 - 4 \cdot -c^2 \cdot 1 = 4c^2 > 0$$

Therefore the wave equation is hyperbolic.

- (b) Given $x = \frac{1}{2}\xi + \frac{1}{2}\eta$ and $t = \frac{1}{2c}\xi - \frac{1}{2c}\eta$

$$u_\xi = u_x x_\xi + u_t t_\xi$$

Question 3